

Modeling air pollution exposure with MATSim: to what extent are complex modeling chains necessary?

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1 Introduction

Exposure to ambient air pollution is a major public health concern, causing an estimated 4.2 million premature deaths annually worldwide, particularly from cardiovascular and respiratory diseases and lung cancer (WHO, 2021). In urban areas, road transport is a major source of air pollutant emissions and contributes to spatially heterogeneous exposure, particularly in densely built environments where pollutant dispersion is strongly influenced by local conditions. Assessing population exposure and its inequalities is therefore essential for informing transport and urban policies. While traditional approaches rely on residential locations, recent studies increasingly account for individual mobility to capture exposure throughout daily activity chains. These approaches combine mobility information with spatially resolved pollutant concentrations derived from measurements, land-use regression, satellite observations, or dispersion and chemistry-transport models (CTMs).

Agent-based models (ABMs) have recently attracted growing attention as a means of representing individual mobility behaviors and their resulting exposure disparities within integrated modeling frameworks. ABMs are well established in transportation research for evaluating mobility policies and are increasingly applied to exposure assessments. These applications range from combining simulated mobility with externally estimated concentration fields to fully integrated modeling chains that explicitly couple mobility, road-traffic emissions, air quality or noise, and exposure models (Gastineau et al., 2023). Such integrated frameworks ensure greater consistency between individual mobility, traffic flows, emissions, pollutant concentrations, and resulting exposures. As integrated modeling chains are becoming increasingly complex, the addition of model components may reduce model incompleteness but also introduce new sources of error and uncertainty that can propagate through the modeling chain and affect exposure estimates. However, the sensitivity of exposure estimates to structural modeling choices across the full modeling chain remains insufficiently explored, particularly regarding their implications for exposure inequalities. Existing scenario-based assessments typically focus on one or two components of the modeling chain, providing limited insight into how alternative structural assumptions across the chain affect exposure estimates and inequality indicators.

This study evaluates the sensitivity of exposure estimates and exposure inequality indicators to key structural assumptions across the full modeling chain, and contributes to ongoing efforts to develop integrated modeling frameworks coupling agent-based mobility models with urban air quality models to assess air pollution exposure in urban areas. The main contribution of this study is to systematically assess the sensitivity of exposure estimates to multiple modeling choices across the modeling chain, including population sample size, static versus dynamic approaches, the spatial resolution of emissions and concentrations in urban areas, and the temporal resolution of exposure. It further assesses exposure using multiple indicators, including population-average exposure, differences across age and income groups, and its spatial distribution.

2 Methodology

This study is based on an open-source modeling chain to assess individuals' exposure with an agent-based approach. We developed a population — mobility — emissions — air quality — exposure modeling chain with an agent-based approach. The population is modeled using eqasim travel demand pipeline (Hörl and Balac, 2021), then mobility is represented using MATSim (Axhausen, Horni, and Nagel, 2016). Emissions from cars are computed based on MATSim outputs with HBEFA traffic-state emission model (Notter et al., 2019), while emissions from other road traffic sources (e.g., from HDV) and other sectors (e.g., heating) come from Airparif emissions inventory. Based on these emissions inputs, land-use data and a meteorological simulation using the Weather Research and Forecasting model (WRF), air quality is represented using a regional chemistry-transport model (kilometer scale), POLAIR 3D, coupled with the street-level urban air quality model MUNICH to determine pollutant concentrations in canyon streets (Kim et al., 2022), both of which integrate the chemistry module SSH-aerosol. This multiscale approach (street-in-grid) offer an accurate representation of the local variability of concentrations related to the aerodynamic effect of buildings and spatial heterogeneity in emissions, while representing the physicochemical generation of secondary aerosols. Finally, we developed a model for personal exposure based on MATSim activity plans, with exposure during each activity or travel relying on the type of streets in which agents are located. When individuals are traveling or located in buildings in wide streets, concentrations from the regional scale model are used, while concentrations from the local scale model are used for buildings along canyon streets. Except for travels by underground (metro, regional railways)¹, concentrations are supposed to be the same as outside in transportation means and indoors.

Table 1: Modeling assumptions of the modeling chain components by scenario

Scenario	Population	Mobility	Emissions	Air quality	Exposure
Reference	25%	Dynamic	Detailed car fleet	Street-level	10 minutes
5% sample	5%	Dynamic	Detailed car fleet	Street-level	10 minutes
10% sample	10%	Dynamic	Detailed car fleet	Street-level	10 minutes
Static	25%	<i>Static</i>	Detailed car fleet	Street-level	10 minutes
Pseudo-dynamic	25%	<i>Pseudo-dynamic</i>	Detailed car fleet	Street-level	10 minutes
Average car	25%	Dynamic	<i>Average car</i>	Street-level	10 minutes
No car	25%	Dynamic	<i>No car</i>	Street-level	10 minutes
Background	25%	Dynamic	Detailed car fleet	<i>Background</i>	10 minutes
Static background	25%	<i>Static</i>	Detailed car fleet	<i>Background</i>	10 minutes
60 minutes	25%	Dynamic	Detailed car fleet	Street-level	<i>60 minutes</i>
30 minutes	25%	Dynamic	Detailed car fleet	Street-level	<i>30 minutes</i>

The case study is the Paris region population, concentrations were simulated for the winter 2018 (8 weeks) for different scenarios, and we assess the exposure to fine particles (PM_{2.5}) and nitrogen dioxide (NO₂), with additional results for black carbon (BC). The different components of the modeling chain were evaluated against observations for the reference scenario, showing satisfying results². Different scenarios are assessed at each stage of the modeling chain (Table 1):

- For the synthetic population, different sample sizes were simulated: 25 % in the reference scenario, which is compared with the 5 % and 10 % scenarios.

¹Concentrations of fine particles in underground public transportation are set to 65.0 µg/m³ based on measurements within stations for the 2018 winter by the Paris public transit operator (RATP).

²For MATSim, we found a 0.9 % error in the distance traveled by cars and 2.2 % in the total number of trips at the regional scale compared with the travel survey. For PM_{2.5} concentrations, MUNICH reproduces well pollution peaks at traffic stations with a fractional bias (FB) of -4%, but Polair3D fails to capture some peaks at background stations resulting in a -43% FB.

- For the representation of mobility, the reference scenario adopts a *dynamic approach* (i.e., using all MATSim events) which is compared with the *static scenario* that consists in approximating exposure by concentrations at the place of residence, and the *pseudo-dynamic approach* in which exposures in all activity places are considered, but the exposure during travel is ignored (i.e., using MATSim activities only).
- In the car emission model, we investigate the representation of the car types in coupled agent-based transportation and emissions models and its impact on simulated concentrations and exposure. The scenario representing the car fleet with detailed car types attributed to households, the *detailed car fleet* scenario, is compared to the scenario where car emissions are modeled using a unique car type representing average emissions, the *average car* scenario, while the scenario with no car emissions is also considered as a reference point in the discussion.
- Regarding the air quality component, we compare the exposure computed using the canyon street model MUNICH nested with the regional CTM Polair3D, that is labeled *street-level* scenario, to the one using only the *background* concentrations simulated with Polair3D.
- In the exposure model, different temporal resolution were compared: a 10 minutes resolution was chosen as reference, but the current practice is to use concentration outputs at an hourly time step, so we compared it with 30 minutes and 60 minutes time steps.

3 Results

For each scenario, we analyzed the mean relative change when comparing the individual exposure between the scenario of interest and the reference and its standard deviation (Table 2), but also the cumulated distribution of individual exposures, the relative changes by age and income group and the maps of mean relative changes in exposure by place of residence.

Table 2: Synthesis of the mean relative change in exposure (RC) and its standard deviation by scenario (σ_{RC}), compared with the reference scenario

Scenario	PM _{2.5} exposure		NO ₂ exposure		BC exposure	
	RC	σ_{RC}	RC	σ_{RC}	RC	σ_{RC}
5% sample	-0.04%	-	+0.03%	-	+0.03%	-
10% sample	-0.05%	-	-0.08%	-	-0.05%	-
Static	-2.5%	8.7%	-2.9%	12.4%	-2.9%	12.4%
Pseudo-dynamic	-1.8%	4.6%	-0.3%	2.8%	-0.4%	2.8%
Average car	+1.2%	1.3%	+3.3%	1.1%	+2.3%	2.1%
No car	-2.8%	3.0%	-13.7%	3.1%	-10.8%	4.8%
Dynamic background	-4.0%	8.0%	-3.7%	7.1%	-4.1%	7.8%
Static background	-5.6%	9.5%	-5.9%	12.3%	-6.0%	10.5%
60 minutes	-8.2%	7.5%	-21.8%	7.3%	-10.0%	8.5%
30 minutes	-1.6%	0.4%	-1.2%	0.7%	-1.7%	0.5%

First, we find important an important underestimation of individual exposure when using only background concentrations from the regional air quality model rather than combining it with the canyon street model (dynamic background scenario), and even more important when also relying on the place of residence rather than MATSim activity plans (static background scenario). These two scenarios lead to a systematic underestimation

of individual $\text{PM}_{2.5}$ exposure (-4% and -5.6%), with substantial variability across individuals and space. This effect is particularly pronounced for children, older adults, and low-income individuals, and is especially relevant in the urban center. These findings suggest that representing concentration heterogeneities in the urban canyon is essential in exposure assessment, especially for Environmental Justice studies and health and economic impact assessments.

Regarding mobility, the static and pseudo-dynamic MATSim scenarios both slightly underestimate population-average exposure to $\text{PM}_{2.5}$ and NO_2 (-2.5% and -2.9%, respectively, for the static scenario). However, the static approach produces substantial spatial variability, particularly for NO_2 exposure in outer suburban areas, as well as differences across income groups. Although average exposure is only moderately affected, these local and subpopulation-level differences indicate that dynamic mobility is preferable for analyses at fine spatial scales and for distributional or economic assessments.

Population sample size has no substantial effect on mean exposure or its overall distribution when comparing 5 %, 10 % and 25 % samples. However, the 5% sample produces substantial local differences in exposure estimates, ranging from -56% to +370%, with smaller but still noticeable effects for the 10% sample. These results suggest that small population samples may be sufficient for population-level exposure estimates but can lead to substantial uncertainty in spatially detailed assessments.

The representation of vehicle types also affects exposure estimates. Assuming an average car, as commonly done in coupled transport-emission models, leads to a slight systematic overestimation of exposure (+1.2% for $\text{PM}_{2.5}$ and +3.3% for NO_2), with limited variation across population groups. However, this effect is substantial relative to the overall contribution of cars to $\text{PM}_{2.5}$ exposure.

Finally, reducing the temporal resolution of the air quality model to 1 hour substantially underestimates personal exposure, with considerable variability across pollutants. In contrast, a 30-minute resolution produces negligible changes in both the population exposure distribution and its fine-scale spatial pattern, suggesting that this level of temporal aggregation may provide a reasonable compromise between model complexity and exposure accuracy.

4 Discussion

This scenario-based analysis highlights the importance of model completeness in MATSim-based air pollution exposure assessment. The lack of representation of concentration heterogeneity within urban street canyons can lead to substantial underestimation of personal exposure, particularly when combined with a static exposure approach. The results also show that population sample size has little influence on population-average exposure, but can strongly affect exposure estimates at the local scale, particularly for small samples. On the contrary, the impact of representing vehicle types using the HBEFA module is too small to justify its additional complexity for exposure assessments. More generally, the analysis highlights the importance of carefully selecting the level of detail of the different components coupled to MATSim according to the study objectives (e.g., health impact assessment or evaluation of transportation policies) and the spatial scale of the exposure assessment.

By considering alternative assumptions across the full modeling chain, this study provides insights into the consequences of commonly used simplifications for exposure assessment. However, the air quality modeling considered only one season, and the relevance of these modeling choices may vary under different atmospheric conditions. Further analyses covering other seasons, particularly summer conditions and ozone exposure, would therefore be needed to assess the generalizability of these findings.

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